Dot Product and Geometry

Recall from last class:

- If A is an $n \times n$ matrix, then $\lambda \in \mathbb{R}$ is an **eigenvalue** of A if there exists a non-zero vector v, called an **eigenvector** corresponding to λ, such that
$$Av = \lambda v.$$
- If λ is an eigenvalue of an $n \times n$ matrix A, then the set of all solutions v to $Av = \lambda v$ is the **eigenspace** of A corresponding to λ.
- **Theorem 1:** The eigenvalues of a triangular matrix are the diagonal entries.
- **Theorem 2:** If $v_1, \dots, v_p$ are eigenvectors corresponding to distinct eigenvalues $\lambda_1, \dots, \lambda_p$ of an $n \times n$ matrix A, then $v_1, \dots, v_p$ are linearly independent.
- If A is an $n \times n$ matrix, then the characteristic polynomial of A, denoted by $p_A(\lambda)$, is given by
$$p_A(\lambda) = \det(A - \lambda I_n).$$
- λ is an eigenvalue of an $n \times n$ matrix A if and only if λ is a zero of the characteristic polynomial.
- If A and B are similar matrices, then they have the same characteristic polynomials and hence the same eigenvalues (with the same algebraic multiplicities).
- An $n \times n$ matrix A is **diagonalizable** if A is similar to a diagonal matrix.
- **Diagonalization Theorem:** An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors. Moreover, $A = PDP^{-1}$ with D diagonal if and only if the columns of P are n linearly independent eigenvectors of A. In this case the diagonal entires of D are the corresponding eigenvalues of A.
- **Theorem 6:** An $n \times n$ matrix A with n distinct eigenvalues is diagonalizable.
- **Theorem 7:** Let A be an $n \times n$ matrix with distinct eigenvalues $\lambda_1, \dots, \lambda_p$.
    1. For $1 \leq k \leq p$, the dimension of the eigenspace of λ_k is at most the algebraic multiplicity of the eigenvalue λ_k.
    2. The matrix A is diagonalizable if and only if the sum of the dimensions of the eigenspaces is n if and only if the characteristic polynomial factors into linear factors and the dimension for each eigenspace is exactly the algebraic multiplicity.
    3. If A is diagonalizable and $\mathcal{B}_k$ is a basis for the eigenspace corresponding to λ_k, then $\mathcal{B}_1, \dots, \mathcal{B}_p$ is an eigenbasis for $\mathbb{R}^n$.

Definition 1. *Let $u, v \in \mathbb{R}^n$, then the* **dot product** *of u and v is given by*

$$u \cdot v = u^T v = u_1 v_1 + \cdots + u_n v_n.$$

Theorem 6.1: Let $u, v, w \in \mathbb{R}^n$ and $c \in \mathbb{R}$.

1. $$u \cdot v = v \cdot u.$$

2. $$(u + v) \cdot w = u \cdot w + v \cdot w.$$

3. $$(cu) \cdot v = c(u \cdot v) = u \cdot (cv).$$

4. $u \cdot u \geq 0$ and $u \cdot u = 0$ if and only if $u = 0$.

Definition 2. *The* **length** *or* **norm** *of the vector v is a non-negative scalar, denoted by $\|v\|$, and defined by*

$$\|v\| = \sqrt{v \cdot v}.$$

Definition 3. *We define the* **distance** *between u and v by*

$$dist(u, v) = \|u - v\|.$$

Definition 4. *The vectors u and v are* **orthogonal** *if $u \cdot v = 0$.*

Theorem 6.2: The vectors u and v are orthongal if and only if

$$\|u\|^2 + \|v\|^2 = \|u + v\|^2.$$

1. Prove the Theorem.

Definition 5. *The set*

$$W^{\perp} = \{z \in \mathbb{R}^n : z \cdot w = 0 \text{ for all } w \in W\}$$

is called the **orthogonal complement** *of W.*

Note: $x \in W^{\perp}$ if and only if x is orthogonal to each vector in a set that spans W. Also $W^{\perp}$ is a subspace of $\mathbb{R}^n$.

Theorem 6.3 Let A be an $m \times n$ matrix, then

$$(\text{Row}(A))^{\perp} = \text{Null}(A) \text{ and } (\text{Col}(A))^{\perp} = \text{Null}(A^T).$$

2. Prove the Theorem.

Cauchy-Schwarz Inequality: For all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ we have

$$|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \, \|\mathbf{y}\|.$$

1. Prove Cauchy-Schwarz.

Note: We can use the dot product to find the angles between vectors; namely

$$u \cdot v = \|u\| \, \|v\| \cos \theta.$$

Triangle Inequality: For all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ we have $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.

2. Prove the triangle inequality.

Definition 6. *A set of vectors* $\{u_1, \ldots, u_p\}$ *in* $\mathbb{R}^n$ *is* **orthogonal** *if* $u_k \cdot u_j = 0$ *for all* $j \neq k$.

Theorem 6.4: If $S = \{u_1, \ldots, u_p\}$ is an orthogonal set of non-zero vectors in $\mathbb{R}^n$, then S is linearly independent and hence a basis for Span S.

3. Prove the Theorem.

Definition 7. *An* **orthogonal basis** *for a subspace* W *of* $\mathbb{R}^n$ *is a basis for* W *that is also an orthogonal set.*

Theorem 6.5: Let $\{u_1, \ldots, u_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. For each $y \in W$, then

$$y = c_1 u_1 + \cdots + c_p u_p$$

where

$$c_j = \frac{y \cdot u_j}{u_j \cdot u_j}.$$

4. Prove the Theorem.

5. Let $u_1 = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$ and $u_2 = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$ and $x = \begin{bmatrix} 9 \\ -7 \end{bmatrix}$. Find x in the $\{u_1, u_2\}$-coordinates.

Dot Product – Answers / Solutions

1. *Proof.* Let $u, v \in \mathbb{R}^n$, then

$$\|u+v\|^2 = (u+v)\cdot(u+v) = u\cdot u + 2u\cdot v + v\cdot v = \|u\|^2 + 2u\cdot v + \|v\|^2 .$$

Thus $\|u+v\|^2 = \|u\|^2 + \|v\|^2$ if and only if $u \cdot v = 0$, which is equivalent to u and v are orthogonal. □

2. *Proof.* We first prove $(\mathrm{Row}(A))^\perp = \mathrm{Null}(A)$. Let $x \in (\mathrm{Row}(A))^\perp$ and let $a_1, \ldots, a_n$ be the rows of A. Since $x \in (\mathrm{Row}(A))^\perp$, we know that $x \cdot a_k$ for all $1 \leq k \leq p$. Now we check if x is in the null space of A:

$$Ax = \begin{bmatrix} a_1 \cdot x \\ \vdots \\ a_n \cdot x \end{bmatrix} = 0.$$

Thus $x \in \mathrm{Null}(A)$.

Suppose $x \in \mathrm{Null}(A)$, then

$$0 = Ax = \begin{bmatrix} a_1 \cdot x \\ \vdots \\ a_n \cdot x \end{bmatrix}.$$

Thus $a_k \cdot x$ =0 for all $1 \leq k \leq n$. Since $a_1, \ldots, a_n$ is a spanning set for the row space of A, we know that x is orhogonal to all of the row space. Thus $x \in \mathrm{Row}(A)^\perp$.

Now we prove $(\mathrm{Col}(A))^\perp = \mathrm{Null}(A^T)$. Recall $\mathrm{Col}(A) = \mathrm{Row}(A^T)$, and hence by the first part of the theorem, we get

$$(\mathrm{Col}(A))^\perp = (\mathrm{Row}(A^T))^\perp = \mathrm{Null}(A^T).$$

□

3. For any scalar t and any $\mathbf{x}, \mathbf{y}$ we have

$$0 \leq \|\mathbf{x} + t\mathbf{y}\|^2 = \|\mathbf{x}\|^2 + 2t\mathbf{x}\cdot\mathbf{y} + t^2\|\mathbf{y}\|^2. \tag{55}$$

The idea is to choose t so as to minimize the expression on the right-hand side of (55). Call the right-hand side $\phi(t)$. It is a quadratic polynomial in t, and it has its minimum when $\phi'(t) = 0$. The corresponding value is when

$$t = \frac{-\mathbf{x}\cdot\mathbf{y}}{\|\mathbf{y}\|^2}.$$

Plugging in this value of t and simplifying gives the Cauchy-Schwarz inequality.

4. Since both sides of the inequality are non-negative, it holds if and only if it holds after squaring both sides. Doing so and cancelling terms shows that it follows from Cauchy-Schwarz.

5. *Proof.* Suppose $c_1u_1 + \cdots + c_pu_p = 0$. Taking the dot product with u_k, we get

$$c_1u_1 \cdot u_k + \cdots + c_pu_p \cdot u_k = 0,$$

and using orthogonality of $u_1, \ldots, u_p$, we get $c_ku_k \cdot u_k = 0$. Now since u_k is non-zero, we get $c_k = 0$. Thus $c_k = 0$ for all $1 \leq k \leq p$, and hence $u_1, \ldots, u_p$ are linearly independent. □

6. *Proof.* Suppose $y = c_1u_1 + \dots c_pu_p$. Now take the dot product of both sides with u_k to get

$$y \cdot u_k = (c_1u_1 + \cdots + c_pu_p) \cdot u_k.$$

Using the orthogonality of $u_1, \dots, u_p$, we can simplify the right hand side to get

$$y \cdot u_k = c_ku_k \cdot u_k.$$

Therefore

$$c_k = \frac{y \cdot u_k}{u_k \cdot u_k}.$$

$\square$

7. In order to find x in the u_1, u_2 coordinates, we need to find scalars c_1, c_2 such that

$$x = c_1u_1 + c_2u_2.$$

Notice that $u_1 \cdot u_2 = 0$, and hence we can use the previous theorem to conclude

$$c_1 = \frac{x \cdot u_1}{u_1 \cdot u_1} = \frac{39}{13} = 3,$$

and

$$c_2 = \frac{x \cdot u_2}{u_2 \cdot u_2} = \frac{26}{52} = \frac{1}{2}.$$

Thus $[x]_{\{u_1,u_1\}} = \begin{bmatrix} 3 \\ \frac{1}{2} \end{bmatrix}$.